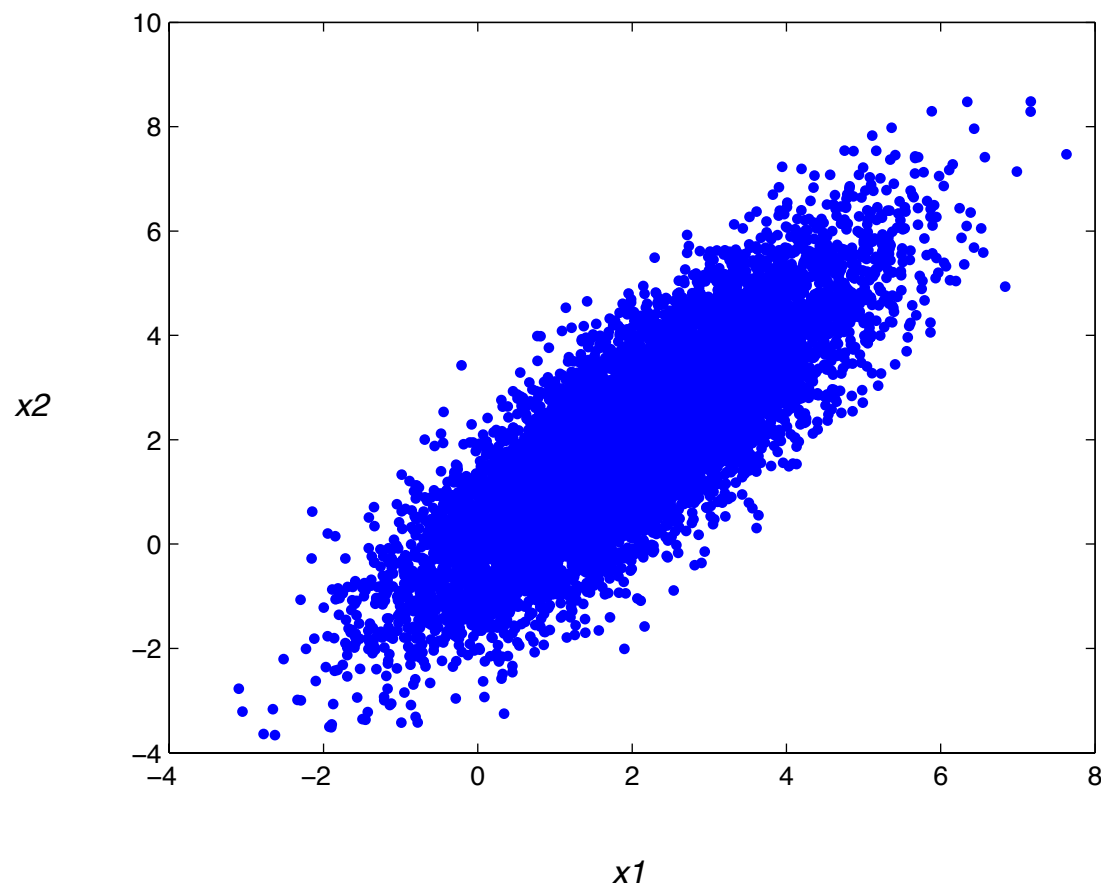


PCA – An Example

$$\mu = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$$

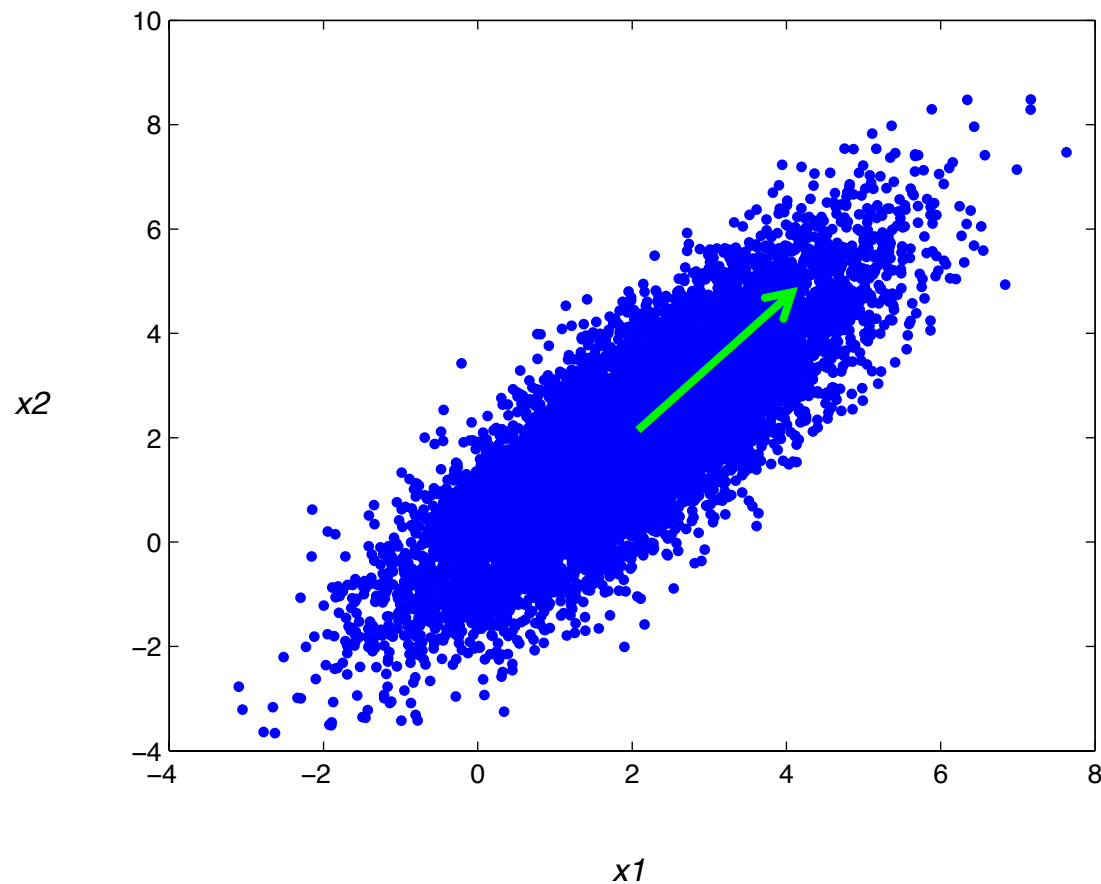
Can you find a vector that would approximate this 2-D space?



PCA – Let's build some intuition

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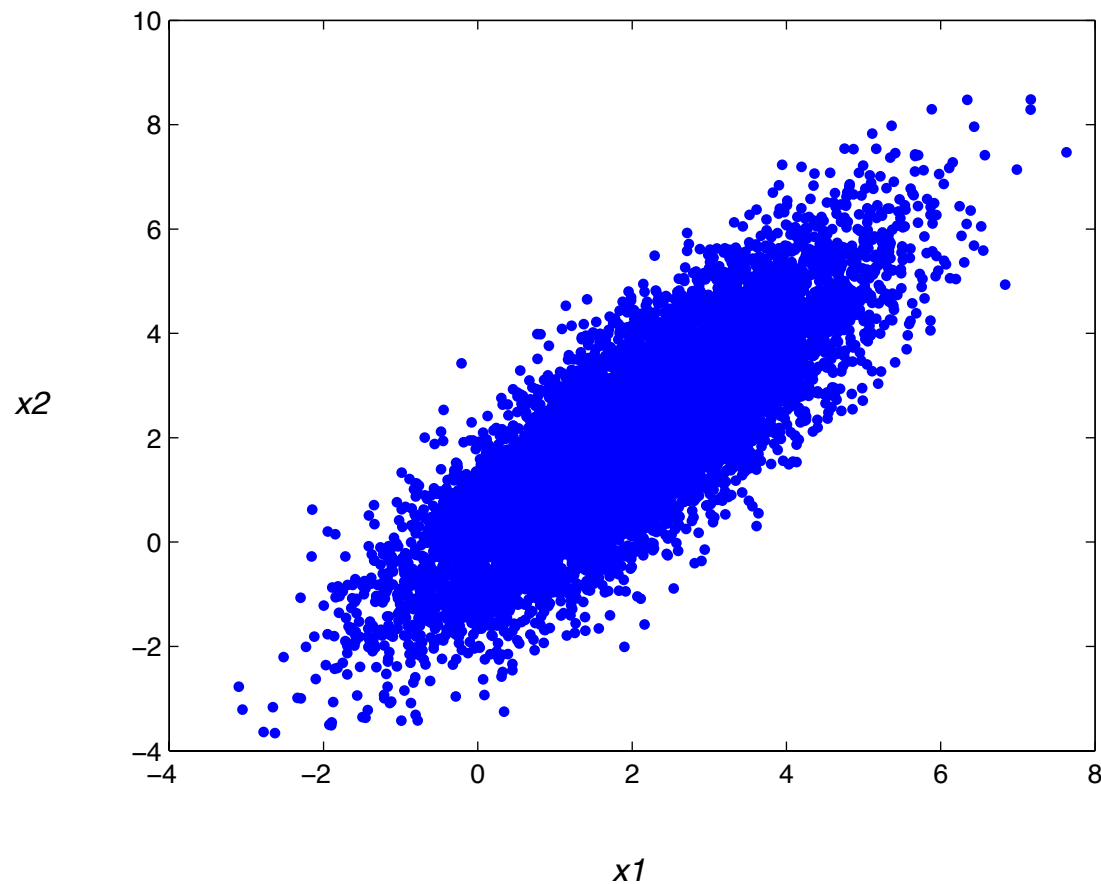
*Can you find a vector that
would approximate this 2-D space?*
Green vector right?



PCA – Let's build some intuition

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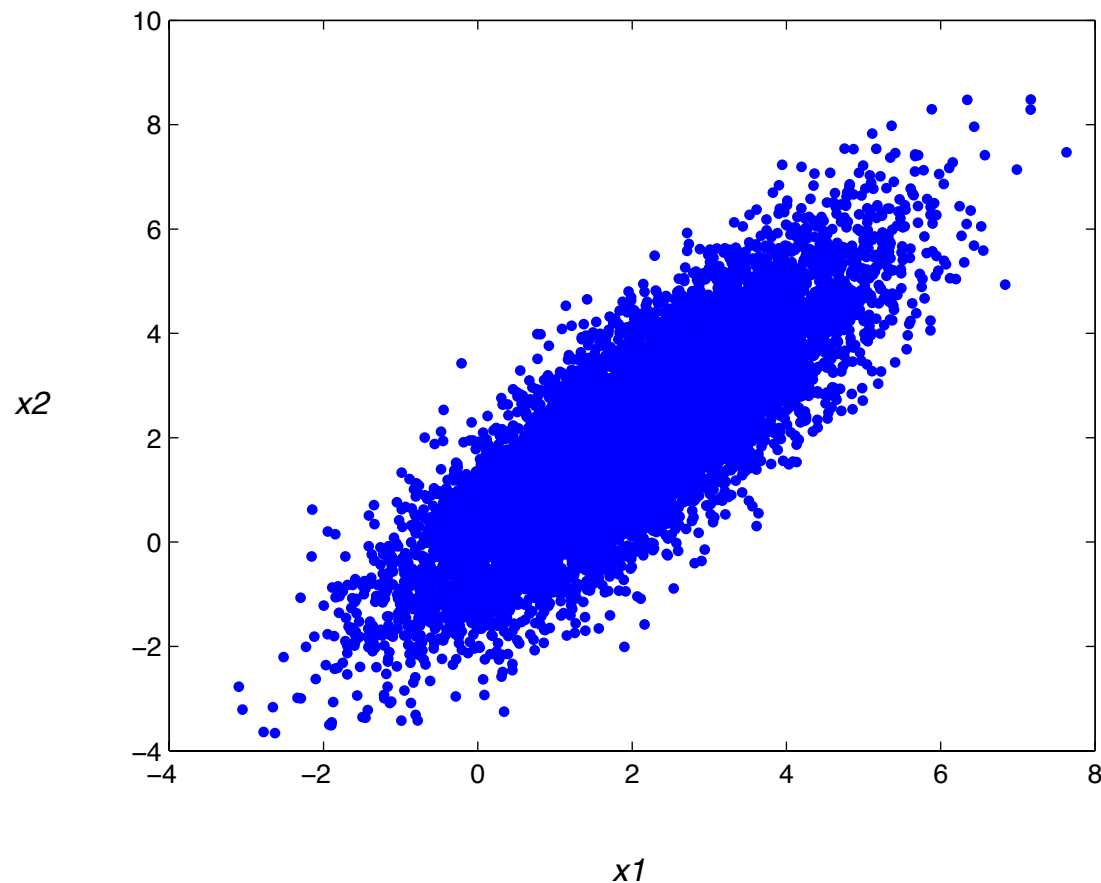
*It would be nice to
diagonalize the covariance matrix
then you have only think about variance*



PCA – Let's build some intuition

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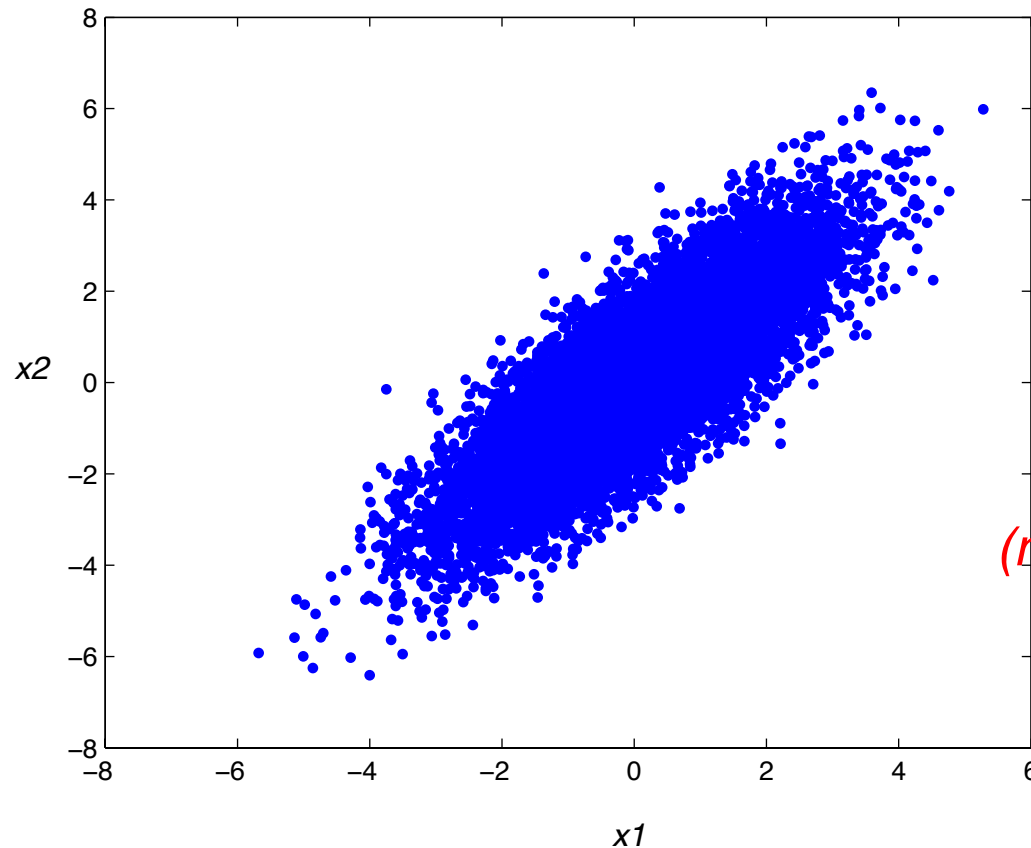
*It would be nice to
diagonalize the covariance matrix
then you have only think about variance
Think eigenvectors of covariance matrix*



PCA – Let's build some intuition

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$$

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Think eigenvectors of covariance matrix*

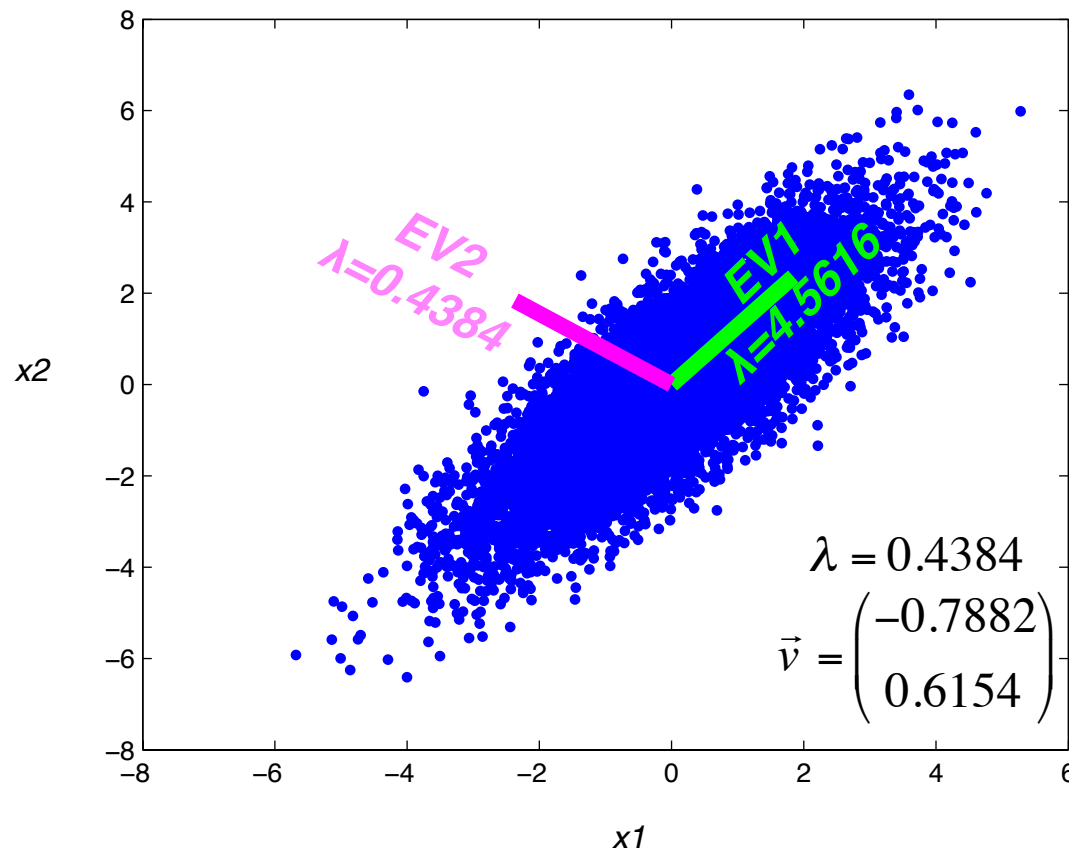


*Let's subtract
the mean first
(makes things simple)*

PCA – Let's build some intuition

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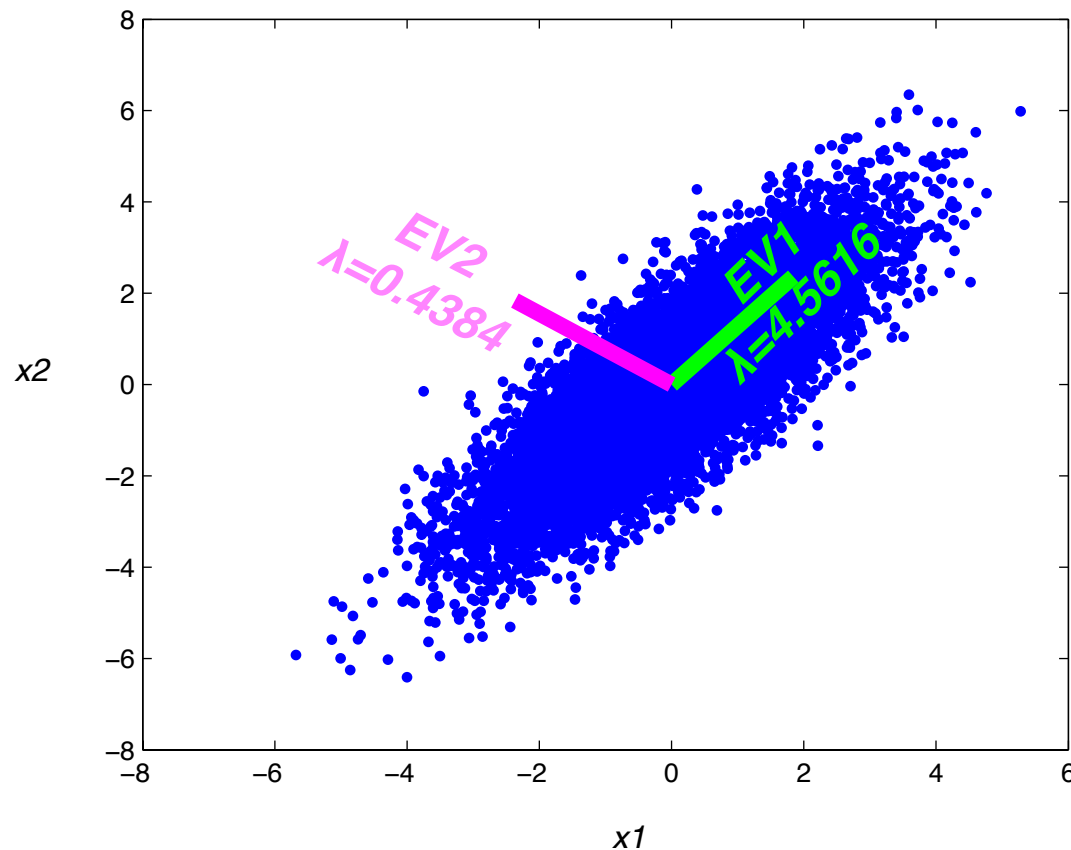
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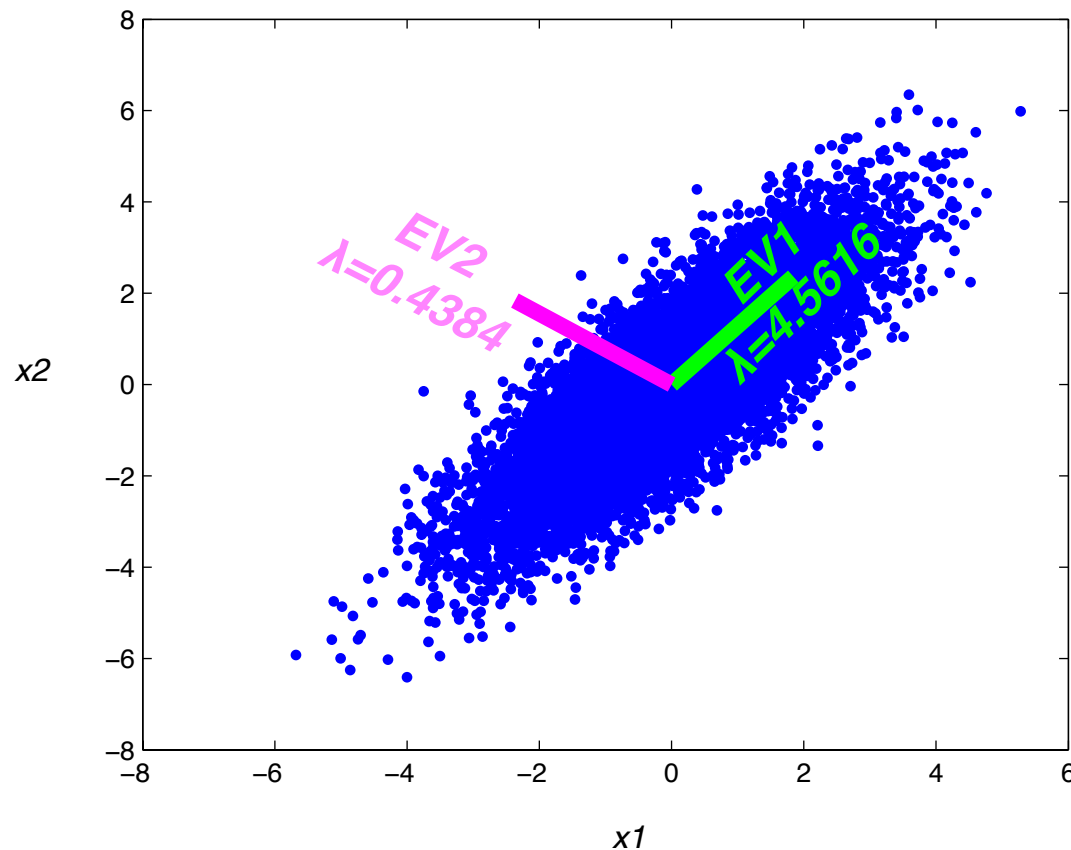
*First eigenvector
accounts for
91.23% of the data*

PCA – Let's build some intuition

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$$

*If I have to pick a vector that
would approximate this 2-D space?*

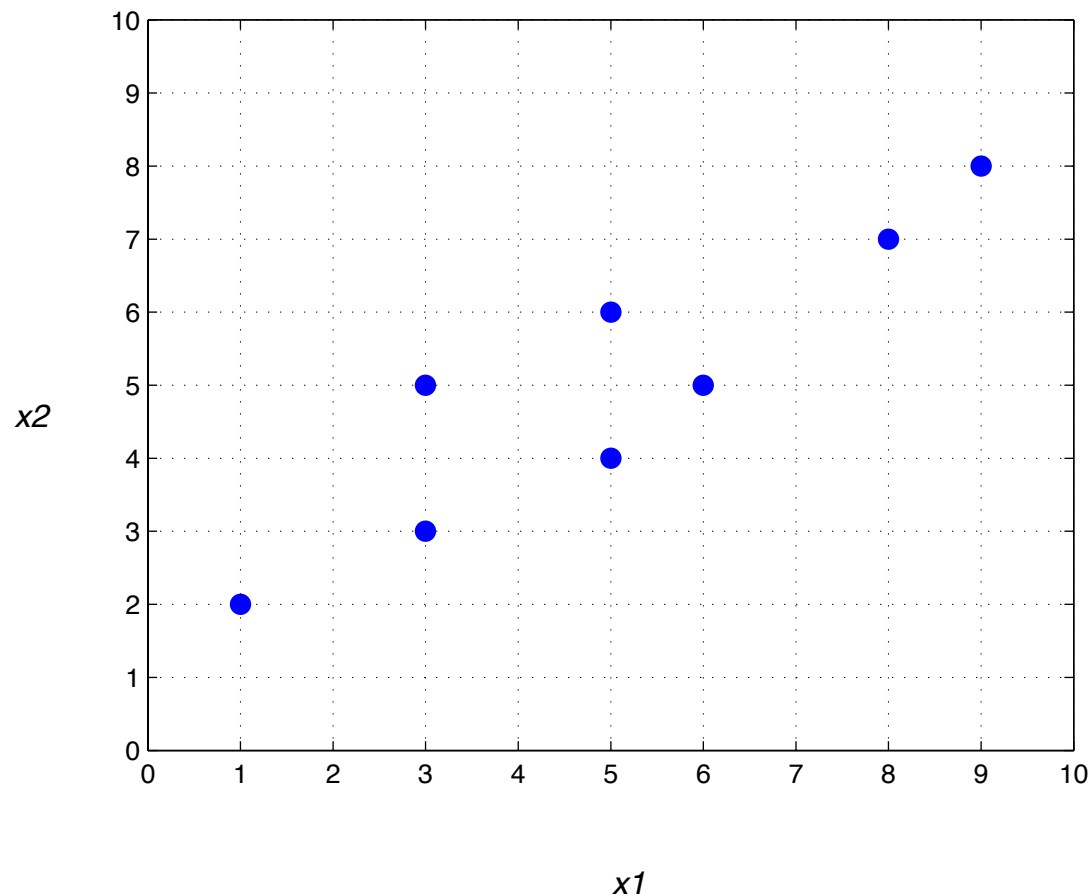
Green vector right



*First eigenvector
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PCA – An Example

- Compute the principal components for the following two-dimensional dataset
 - $X=(x_1,x_2)=\{(1,2),(3,3),(3,5),(5,4),(5,6),(6,5),(8,7),(9,8)\}$

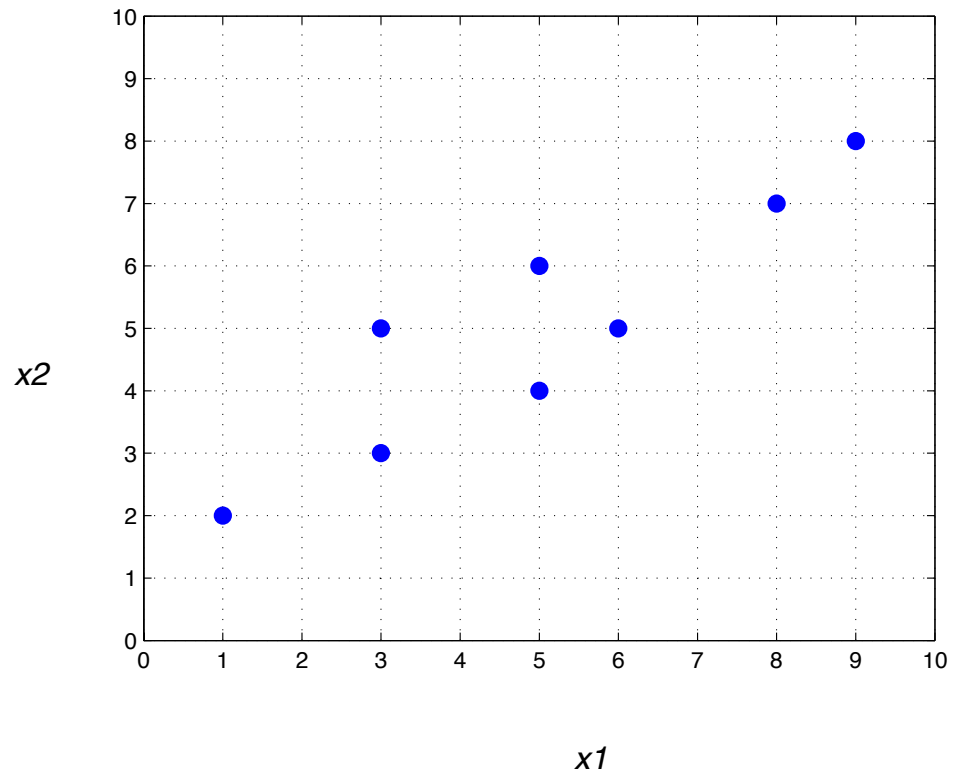


PCA – An Example

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- $X=(x_1,x_2)=\{(1,2),(3,3),(3,5),(5,4),(5,6),(6,5),(8,7),(9,8)\}$

*Step 1: Determine the
Sample Covariance Matrix*



PCA – An Example

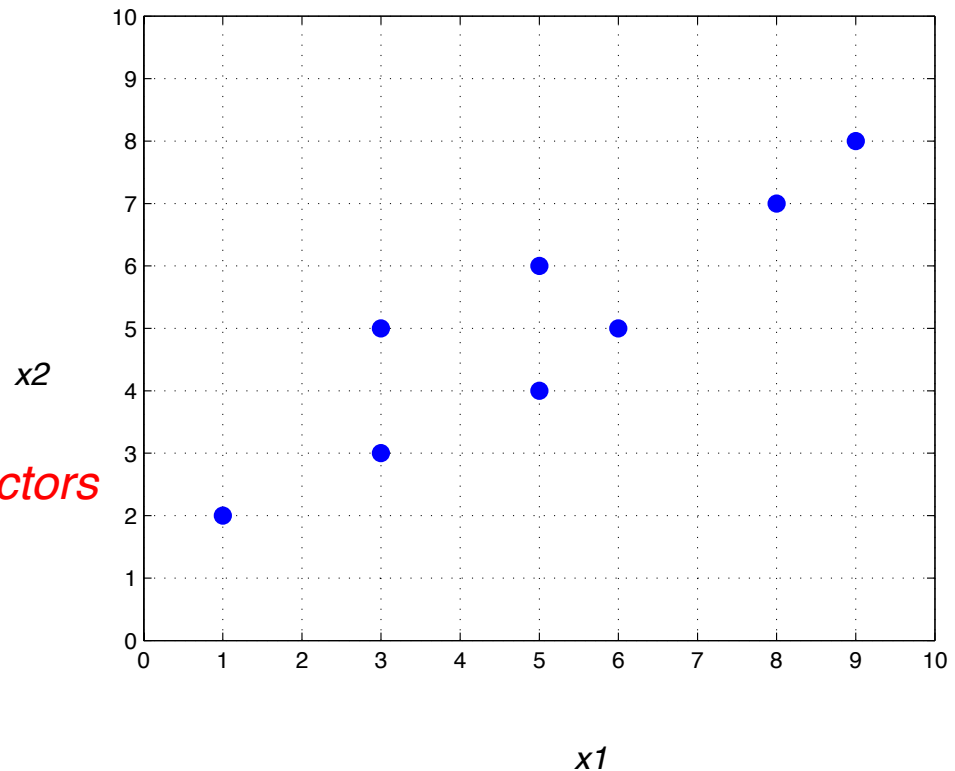
- Compute the principal components for the following two-dimensional dataset

- $X=(x_1,x_2)=\{(1,2),(3,3),(3,5),(5,4),(5,6),(6,5),(8,7),(9,8)\}$

Step 1: Determine the Sample Covariance Matrix

$$\Sigma = \begin{pmatrix} 6.25 & 4.25 \\ 4.25 & 3.5 \end{pmatrix}$$

Step 2: Find eigenvalues and eigenvectors of the covariance matrix



PCA – An Example

- Compute the principal components for the following two-dimensional dataset

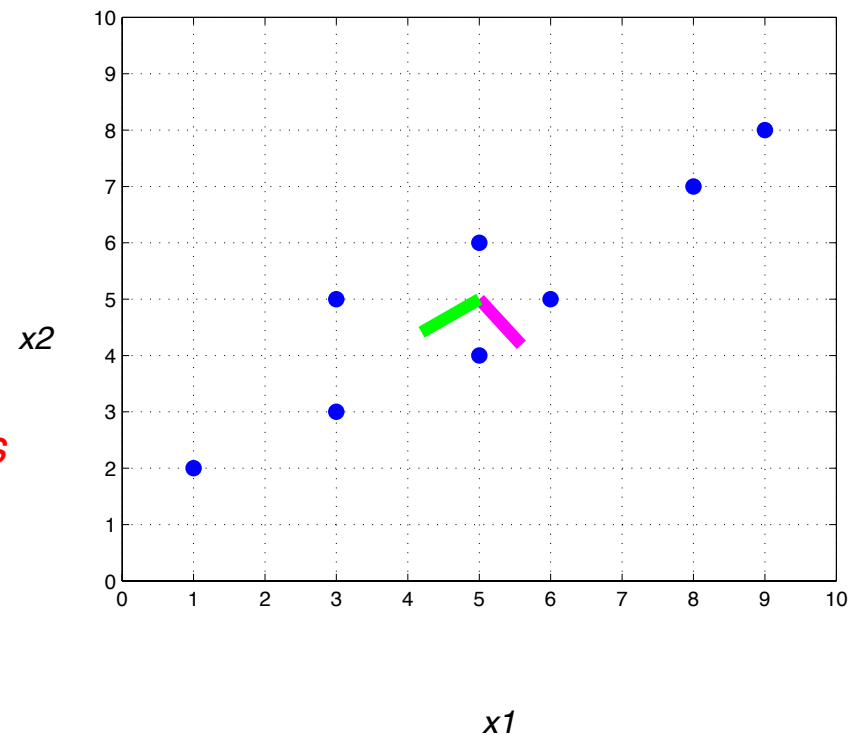
- $X=(x_1,x_2)=\{(1,2),(3,3),(3,5),(5,4),(5,6),(6,5),(8,7),(9,8)\}$

Step 1: Determine Covariance Matrix

$$\Sigma = \begin{pmatrix} 6.25 & 4.25 \\ 4.25 & 3.5 \end{pmatrix}$$

Step 2: Find eigenvalues and eigenvectors of the covariance matrix

$$\begin{aligned} \lambda &= 0.4081 & \lambda &= 9.3419 \\ \vec{v} &= \begin{pmatrix} 0.5883 \\ -0.8086 \end{pmatrix} & \vec{v} &= \begin{pmatrix} -0.5883 \\ -0.8086 \end{pmatrix} \end{aligned}$$



Now the Math Part

- **How did we decide that eigenvectors of the covariance matrix will retain maximum information?**
- **What are the assumptions behind PCA**
 - First, the data distribution is unimodal Gaussian in nature
 - Fully explained by the first two moments of the distribution i.e. mean and covariance
 - Information is in the variance
- **PCA dimensionality reduction**
 - The optimal* approximation of a random vector $x \in \mathbb{R}^N$ by a linear combination of M ($M < N$) independent vectors is obtained by projecting the random vector x onto the eigenvectors φ_i corresponding to the largest eigenvalues λ_i of the covariance matrix Σ_x

(*optimality is defined as the minimum of the sum-square magnitude of the approximation error)

Lets do the derivation

- **The objective of PCA is to perform dimensionality reduction while preserving as much of the randomness (variance) in the high-dimensional space as possible**

- Let $\mathbf{x}$ be an N -dimensional random vector, represented as a linear combination of orthonormal basis vectors $[\varphi_1 | \varphi_2 | \dots | \varphi_N]$ as

$$\vec{x} = \sum_{i=1}^N y_i \varphi_i \quad \text{where} \quad \varphi_i \perp \varphi_j$$

- Where y_i is the weighting coefficient of basis vector formed by taking the inner product of $\mathbf{x}$ with φ_i :

$$y_i = \vec{x}^T \varphi_i$$

Lets do the derivation

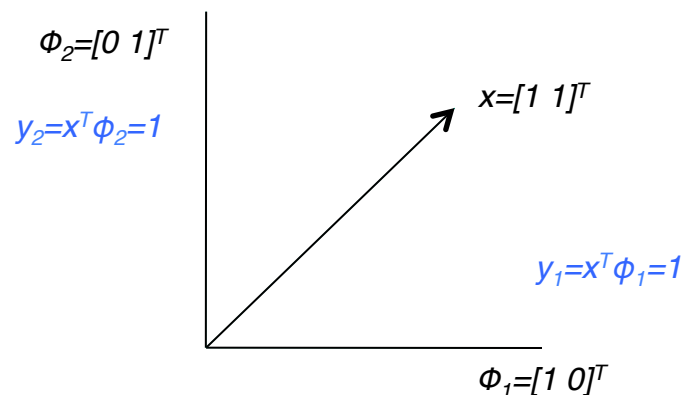
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$$y_i = \vec{x}^T \varphi_i$$



$$\vec{x} = y_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Lets do the derivation

- Suppose we choose to represent $\mathbf{x}$ with only M ($M < N$) of the basis vectors. We can do this by replacing the components $[y_{M+1}, \dots, y_N]^T$ with some pre-selected constants b_i

$$\hat{x}(M) = \sum_{i=1}^M y_i \varphi_i + \sum_{i=M+1}^N b_i \varphi_i$$

- The representation error is:

Lets do the derivation

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$$\hat{x}(M) = \sum_{i=1}^M y_i \varphi_i + \sum_{i=M+1}^N b_i \varphi_i$$

- The representation error is

$$\begin{aligned} \Delta x(M) &= \vec{x} - \hat{x}(M) \\ &= \sum_{i=1}^M y_i \varphi_i + \sum_{i=M+1}^N y_i \varphi_i - \left(\sum_{i=1}^M y_i \varphi_i + \sum_{i=M+1}^N b_i \varphi_i \right) \\ &= \sum_{i=M+1}^N y_i \varphi_i - \sum_{i=M+1}^N b_i \varphi_i \\ &= \sum_{i=M+1}^N (y_i - b_i) \varphi_i \end{aligned}$$

Lets do the derivation

- We can measure this representation error by the mean-squared magnitude of Δx
- Our goal is to find the basis vectors φ_i and constants b_i that minimize this mean-square error

$$\begin{aligned} E[|\Delta x(M)|^2] &= \frac{1}{S} \sum_{i=1}^S \bar{\varepsilon}_i^2(M) \quad S = \text{samples} \\ &= E \left[\left(\sum_{i=M+1}^N (y_i - b_i) \varphi_i \right) \left(\sum_{i=M+1}^N (y_i - b_i) \varphi_i \right) \right] \\ &= E \left[\left(\sum_{i=M+1}^N \sum_{j=M+1}^N (y_i - b_i)(y_j - b_j) \varphi_i^T \varphi_j \right) \right] \\ &= \sum_{i=M+1}^N E[(y_i - b_i)^2] \quad \begin{bmatrix} \varphi_i \perp \varphi_j \\ |\varphi_i| = 1 \end{bmatrix} \end{aligned}$$

Lets do the derivation

■ Find b_i

- The optimal values of b_i can be found by computing the partial derivative of the objective function and equating it to zero

$$\frac{\partial}{\partial b_i} \left(\sum_{i=M+1}^N E[(y_i - b_i)^2] \right) = -2(E[y_i - b_i]) = 0$$

$$= -2(E[y_i] - b_i) = 0$$

*$E[x]$ is a linear operator
 $E[A-B]=E[A]-E[B]$*

$$b_i = E[y_i]$$

- Intuitive: replace the discarded dimensions y_i 's by their expected value (or mean)

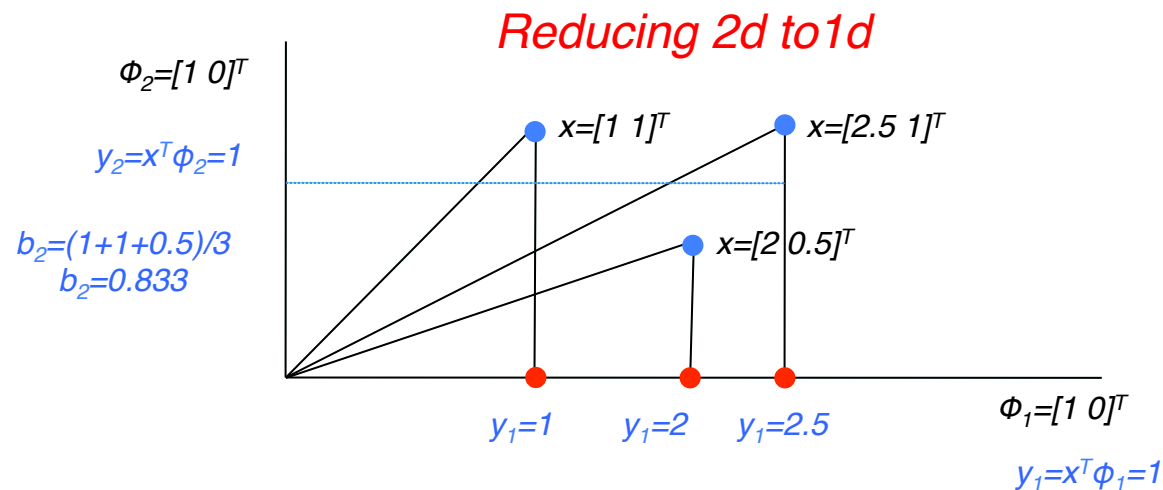
Lets do the derivation

■ Find b_i

- As we have done earlier in the course, the optimal values of b_i can be found by computing the partial derivative of the objective function and equating it to zero

$$b_i = E[y_i]$$

- Intuitive: replace the discarded dimensions y_i 's by their expected value



Lets do the derivation

- The Mean-Squared-Error can now be written as

$$\begin{aligned}\bar{\varepsilon}^2(M) &= \sum_{i=M+1}^N E\left[(y_i - b_i)^2\right] \\ &= \sum_{i=M+1}^N E\left[(y_i - E[y_i])^2\right]\end{aligned}$$

- where

$$y_i = \vec{x}^T \varphi_i$$

Lets do the derivation

- The Mean-Squared-Error can now be written as

$$\bar{\varepsilon}^2(M) = \sum_{i=M+1}^N E \left[\left(y_i - E[y_i] \right)^2 \right]$$

where

$$y_i = \vec{x}^T \varphi_i$$

- Substituting y_i in the MSE equation we get:

$$\begin{aligned} \bar{\varepsilon}^2(M) &= \sum_{i=M+1}^N E \left[\left(x^T \varphi_i - E[x^T \varphi_i] \right)^2 \right] \\ &= \sum_{i=M+1}^N \varphi_i^T \underbrace{E \left[\left(x - E[x] \right) \left(x - E[x] \right)^T \right]}_{\substack{\text{data} \quad \text{covariance}}} \varphi_i \\ &= \sum_{i=M+1}^N \varphi_i^T \Sigma_x \varphi_i \end{aligned}$$

Where Σ_x is the covariance matrix of x

Lets do the derivation

- We seek to find the solution that minimizes the MSE and is also subject to the normality constraint ($\phi^T \phi = 1$), which we incorporate into the expression using a set of Lagrange multipliers λ_i

$$\bar{\varepsilon}^2(M) = \sum_{i=M+1}^N \varphi_i^T \Sigma_x \varphi_i + \sum_{i=M+1}^N \lambda_i (1 - \varphi_i^T \varphi_i)$$

- Computing partial derivatives with respect to ϕ_i

$$\frac{d}{d\varphi_i} \left(\sum_{i=M+1}^N \varphi_i^T \Sigma_x \varphi_i + \sum_{i=M+1}^N \lambda_i (1 - \varphi_i^T \varphi_i) \right) = 2(\Sigma_x \varphi_i - \lambda_i \varphi_i) = 0$$

$$\text{Note: } \frac{d}{dx} (x^T A x) = (A + A^T) x \stackrel{A \text{ symmertric}}{=} 2Ax$$

$$\Rightarrow \underbrace{\Sigma_x \varphi_i}_{\text{Eigenvalue}} = \underbrace{\lambda_i \varphi_i}_{\text{problem}}$$

- So φ_i and λ_i are the eigenvectors and eigenvalues of the covariance matrix Σ_x

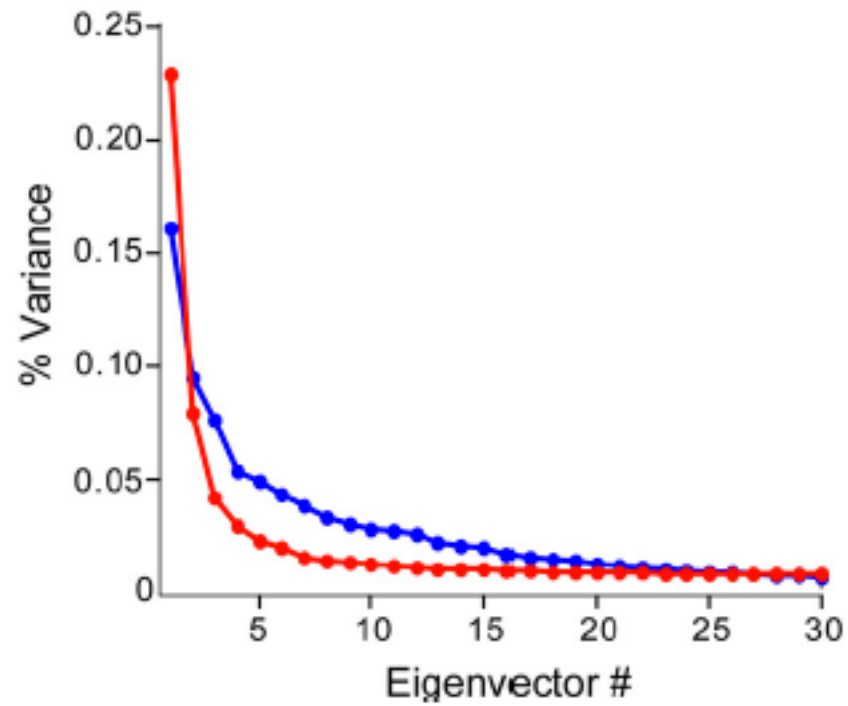
Lets do the derivation

- We can express the sum-squared error as

$$\bar{\varepsilon}^2(M) = \sum_{i=M+1}^N \varphi_i^T \Sigma_x \varphi_i = \sum_{i=M+1}^N \varphi_i^T \lambda_i \varphi_i = \sum_{i=M+1}^N \lambda_i$$

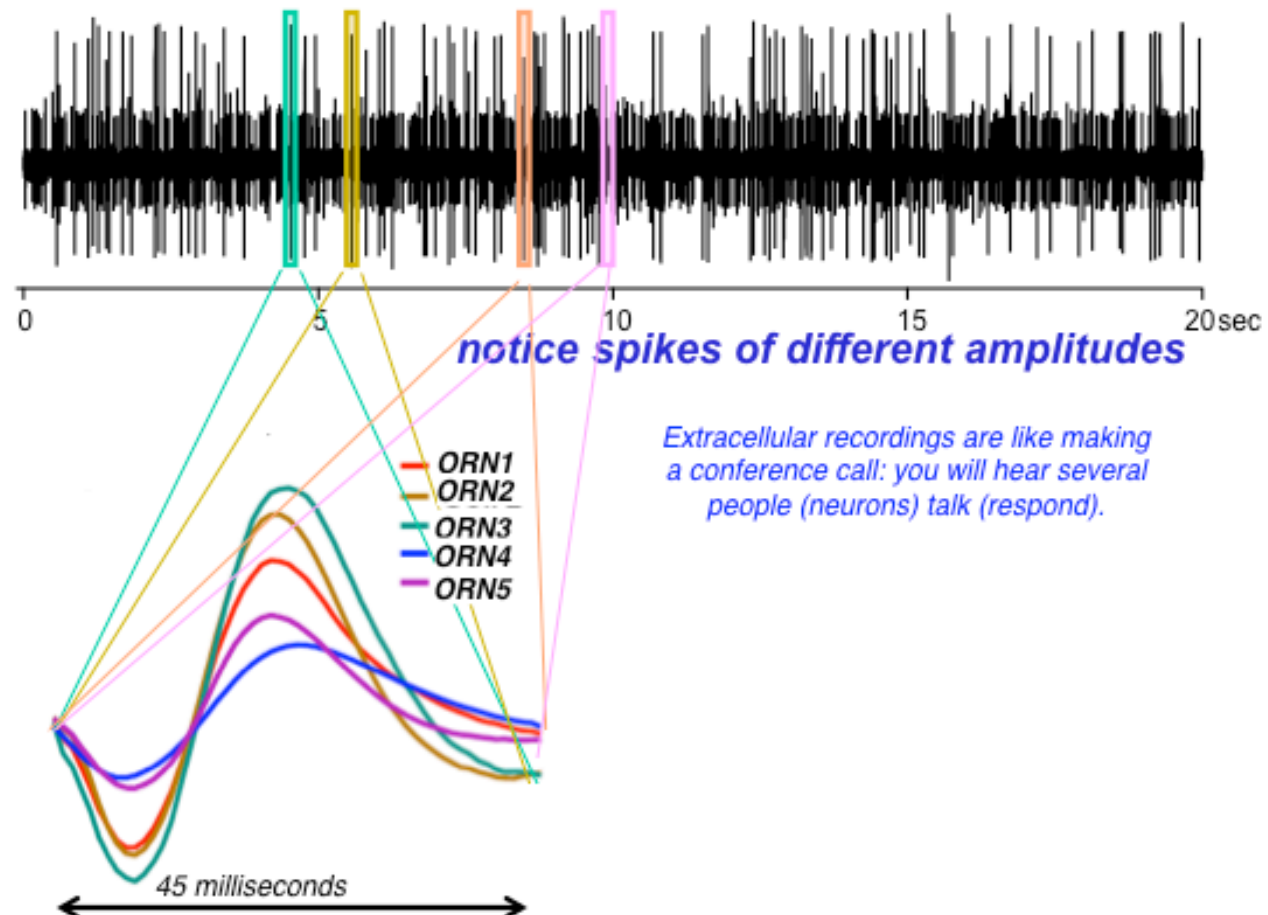
- In order to minimize this measure, λ_i will have to be smallest eigenvalues
 - Therefore, to represent x with minimum sum-square error, we will choose the eigenvectors φ_i corresponding to the largest eigenvalues λ_i

Scree plot



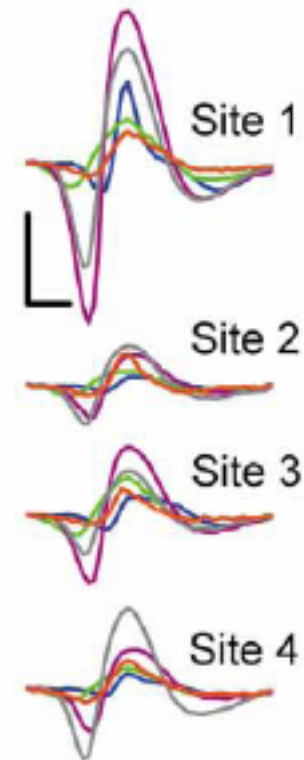
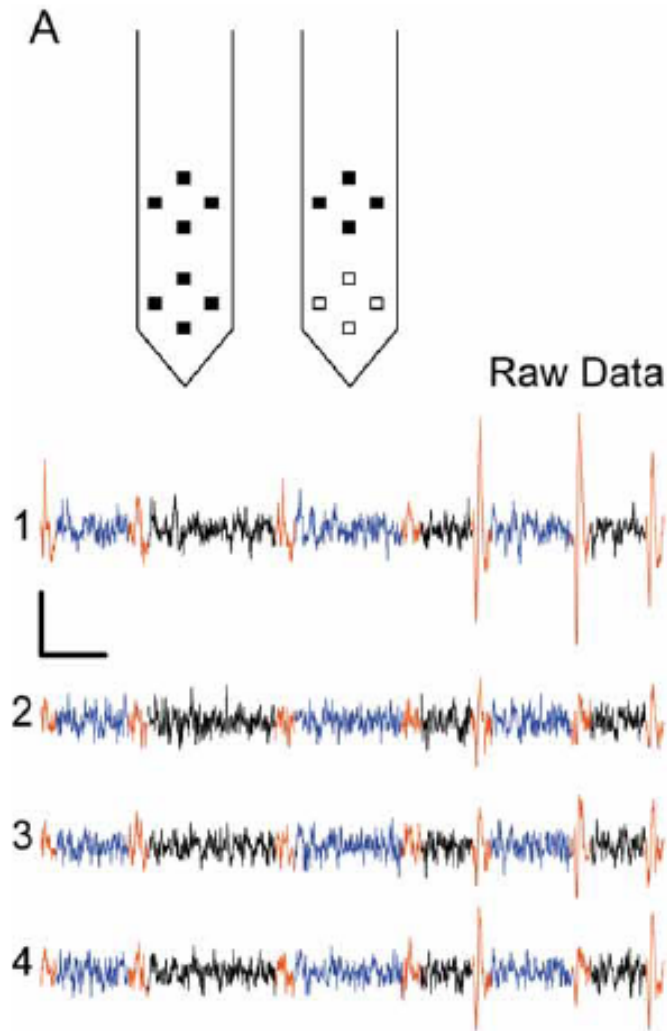
Applications of PCA: Spike Sorting

■ Extracellular recording (single electrode)



Applications of PCA: Spike Sorting

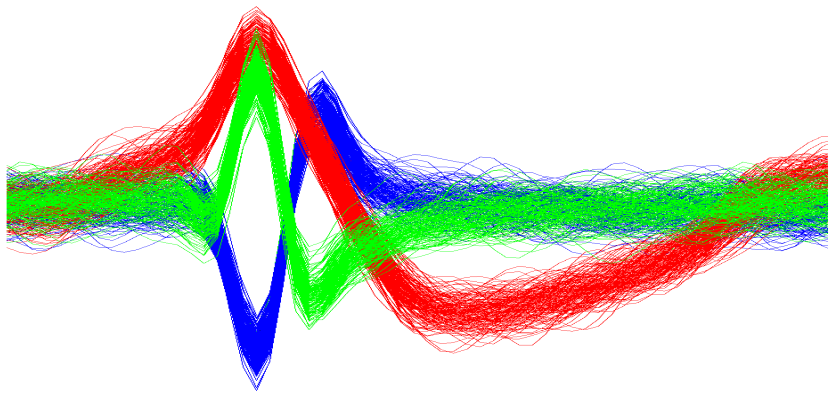
■ Multi-unit recording



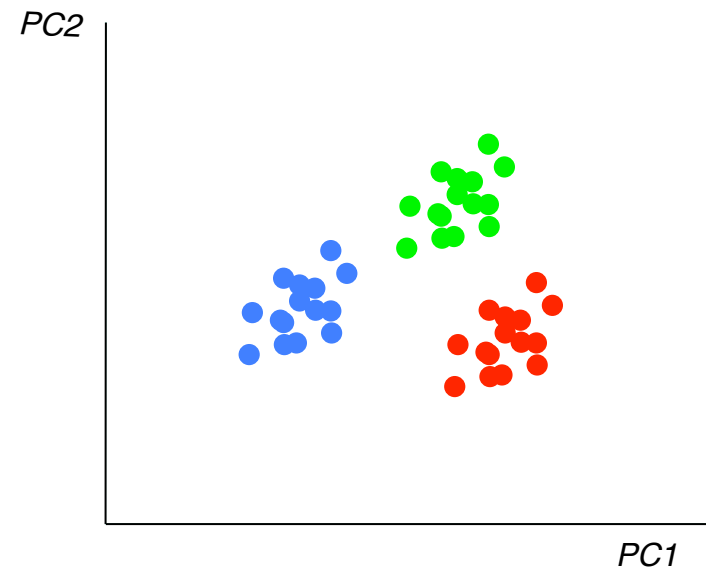
*Notice different colors (represent different neurons) have different patterns of extracellular activity)
[from Pouzat et al. 2002]*

Applications of PCA: Spike Sorting

■ Multi-unit recording



Different colors (represent different neurons) have different patterns of extracellular activity)



Clustering after PCA allows identification of events from a single source (neuron)

Applications of PCA: Neural Coding

- **Information is represented by spiking patterns of neural ensembles**
- **Visualization of high-dimensional neural activity**